ETH zürich

Mobile Computing

Mobile A.I. with Local Large Language Models

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Course Outline

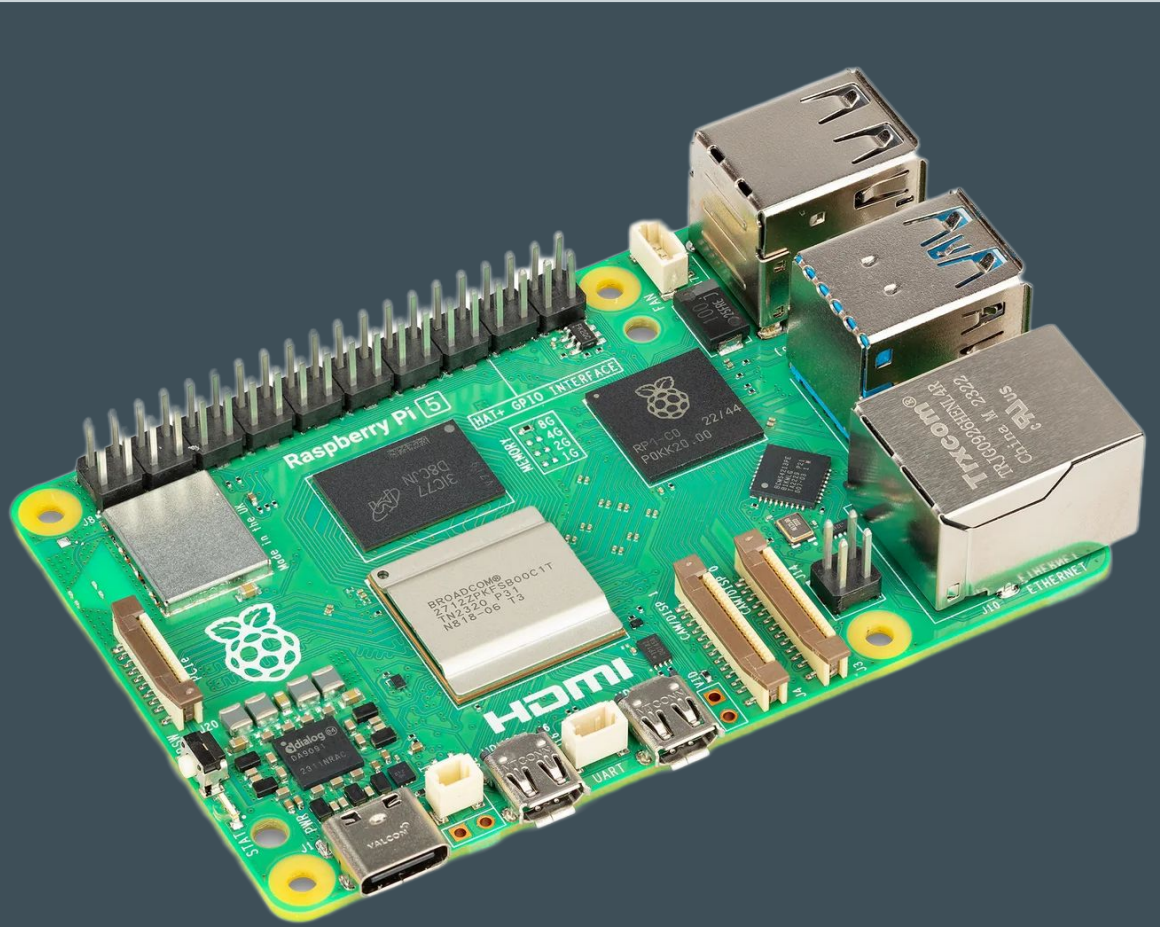
Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

Goals - What we want to achieve

- Build intuition about how LLMs operate in resource-constrained environments, and how to guide them effectively
  - Understand the core principle of text prediction in Local LLMs, and how prompt design influences the model's output
  - Experiment with generation parameters (e.g., temperature, max tokens)
  - Compare Local LLM behaviors on different tasks: decoding, reasoning
  - Reflect on the capabilities and limitations of Local LLMs, for example accuracy, and creativity


Raspberry Pi 5
HAT+ GPIO INTERFACE
BROADCOM
HDMI
UART
ETHERNET

From Cloud to Local LLMs

- Large Language Models (LLMs) usually require powerful, cloud-based infrastructure
- Local LLMs (LLLMs) make it possible to run models directly on personal or edge devices
- This enables
  - Greater control over data and computation
  - Offline operation and reduced dependency on external services

Local vs. Cloud-Based LLMs

- Local LLMs
  - Full control over data privacy and security
  - Offline functionality (no internet dependency)
  - Lower operational costs for small-scale use
- Cloud-Based LLMs
  - Access to cutting-edge, large-scale models
  - High computational performance and scalability
  - Continuous updates and fine-tuning from providers
- Trade-offs
  - Local = privacy and autonomy, but limited power
  - Cloud = performance and convenience, but less control

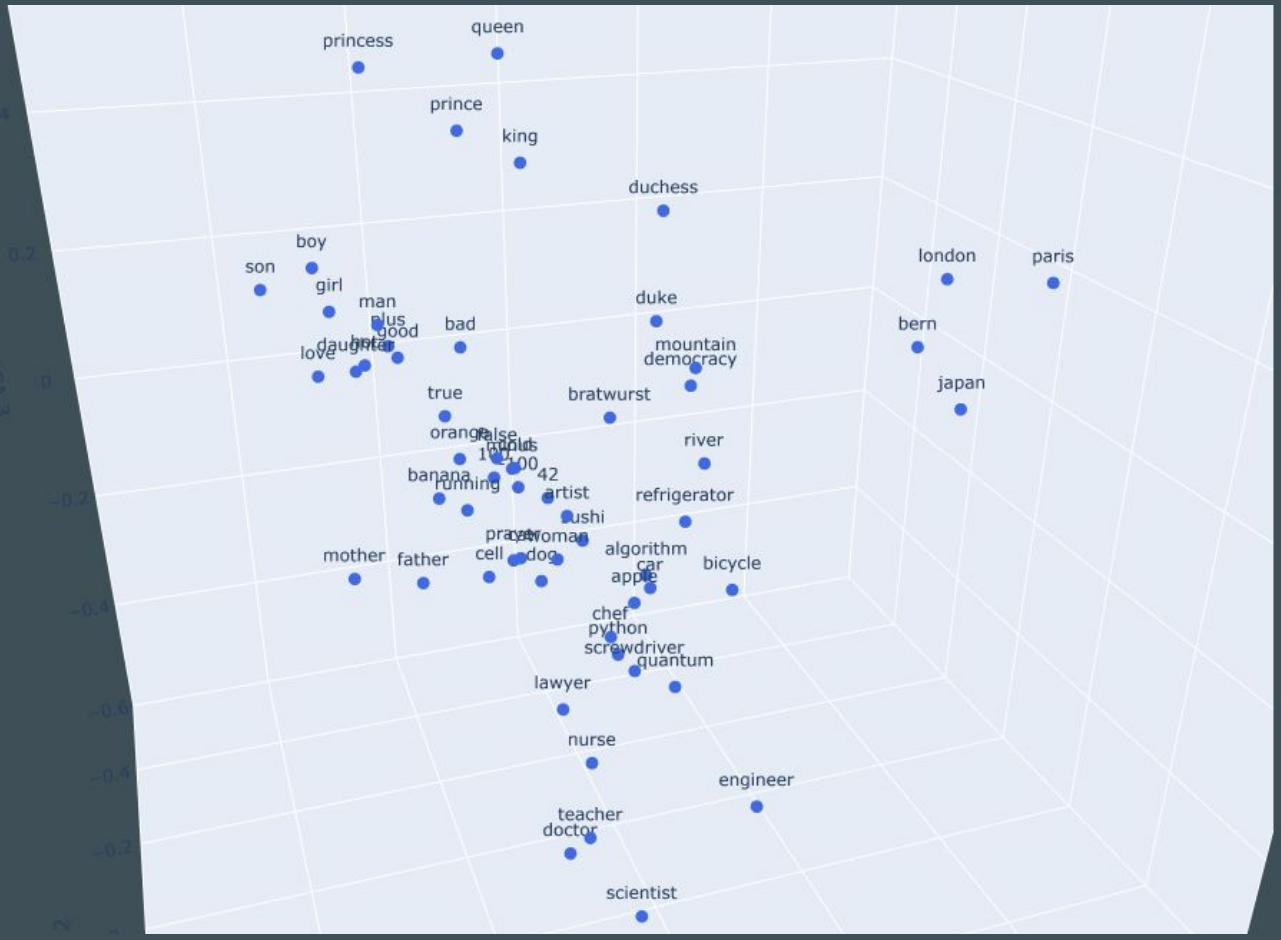
What are Large Language Models (LLMs)?

- AI systems that process and generate text
  - Transform input sequence (question, instruction, statement) into output sequence (continuation, answer, reformulation)
  - Work token-by-token, predicting the next most likely word
  - Evolved from narrow-task AI (e.g., automated translation) to general-purpose models that learn text patterns across massive datasets
- Applications include translation, code generation, creative writing, Q&A, and summarization

What Are Embeddings?

- Embeddings are numerical encodings of words, sentences, or documents
- They transform language into continuous vector spaces
- Similar words lie close together in this space
- Distance and direction between vectors encode:
  - Similarity (e.g., synonyms)
  - Relationships (e.g., gender, roles, analogies)

Some Word Embeddings in “all-MiniLM-L6-v2”



Exercise: 05.01 Understanding Tokenization

- Open in your course folder: 02_Notebooks/05_Mobile_AI/05.01_Understanding_Tokenization.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.02 Understanding Word Embedding

- Open in your course folder:
  02_Notebooks/05_Mobile_AI/
  05.02_Understanding_Word_Embedding.ipynb
- Run the cells and answer the questions.
- Save the completed notebook.

Exercise: 05.03 Understanding Attention

- Open in your course folder: 02_Notebooks/05_Mobile_AI/05.03_Understanding_Attention.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.04 Getting Started with Raspberry Pi

- Open in your course folder:
  02_Notebooks/05_Mobile_AI/
  05.04_GettingStarted_with_Raspberry_Pi.ipynb
- Run the cells and answer the questions.
- Save the completed notebook.

Exercise: 05.05 Introduction to LLMs

- Open in your course folder: 02_Notebooks/05_Mobile_AI/ 05.05_Introduction_to_LLMs.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.06 Understanding Training

- Open in your course folder: 02_Notebooks/05_Mobile_AI/ 05.06_Understanding_Training.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.07 Introduction to Local LLMs

- Open in your course folder:
  02_Notebooks/05_Mobile_AI/
  05.07_Introduction_to_Local_LLMs.ipynb
- Run the cells and answer the questions.
- Save the completed notebook.

Exercise: 05.08 Hardware for Local LLMs

- Open in your course folder: 02_Notebooks/05_Mobile_AI/05.08_Hardware_for_Local_LLMs.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.09 Making Models Smaller

- Open in your course folder: 02_Notebooks/05_Mobile_AI/05.09_Making_Models_Smaller.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.10 PID meets AI

- Open in your course folder:
  02_Notebooks/05_Mobile_AI/
  05.10_PID_meets_AI.ipynb
- Run the cells and answer the questions.
- Save the completed notebook.

Exercise: 05.11 Local RAG meets Tit-for-Tat

- Open in your course folder:
  02_Notebooks/05_Mobile_AI/
  05.11_Local_RAG_meets_Tit-for-Tat.ipynb
- Run the cells and answer the questions.
- Save the completed notebook.

Exercise: 05.12 Understanding Inference Control

- Open in your course folder:
  02_Notebooks/05_Mobile_AI/
  05.12_Understanding_Inference_Control.ipynb
- Run the cells and answer the questions.
- Save the completed notebook.

Exercise: 05.13 Experimenting with Prompts

- Open in your course folder: 02_Notebooks/05_Mobile_AI/05.13_Experimenting_with_Prompts.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.14 New Kinds of Language Models

- Open in your course folder: 02_Notebooks/05_Mobile_AI/05.14_New_Kinds_of_Language_Models.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.15 Local Agents I

- Open in your course folder: 02_Notebooks/05_Mobile_AI/05.15_Local_Agents_I.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.16 Local Model Context Protocol

- Open in your course folder: 02_Notebooks/05_Mobile_AI/05.16_Local_Model_Context_Protocol.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.17 Local Agents II: Vibe Coding

- Open in your course folder: 02_Notebooks/05_Mobile_AI/05.17_Local_Agents_II.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.18 Build Your Own Model

- Open in your course folder: 02_Notebooks/05_Mobile_AI/05.18_Build_Your_Own_Model.ipynb
- Work through the notebook and complete the exercise.
- Save the completed notebook and upload it to Moodle.

Exercise: 05.19 Know When You Don't Know

- Open in your course folder: 02_Notebooks/05_Mobile_AI/05.19_GRAND_FINALE_Know_When_You_Dont_Know.ipynb
- Read the competition and submission instructions.
- Build a quiz player that knows when to answer or pass.

